CSC2001F – Database Assignment 3 – 2025

Approximate mark allocation for each question is shown in square brackets.

This assignment is done in your group of 5 students. Only ONE student must submit your work.

1) Consider the information that is kept in Amathuba, and draw an Entity-Relationship (ER) model for this data – NB This is a model of the data, not of a database, so do NOT submit a relational database schema diagram. Use the ER notation exactly as used in lectures. Marks will be given for **accuracy** and for the **extent** to which you appropriately use the power/richness of this ER notation. You can omit less important parts of the Amathuba system if they will not require using any ER notation not already used in your model. You can hand-draw, or use a free tool for ER diagram creation, as you prefer. [18]

2) Give an appropriate relational database design for this model. Just write the name of each relation followed by the names of its attributes in brackets; underline the attributes in each primary key, and use capital letters (uppercase) to indicate all foreign keys. Example: CrazyTable (Name, DiscussionPost, GRADEITEM) is a (terrible) relation with 3 columns; primary key is Name and GRADEITEM is a foreign key. [11]

3) Give any one functional dependency that holds for the attributes in this data. [1]

4) Give any one transitive dependency that holds for the attributes in this data [1]

5) Give any one partial functional dependency that holds for the attributes in this data. [1]

6) Choose any 1 relation in your answer to (b) above that represents a many-to-many relationship, and state why it is, or is not, in 3rd normal form. If you refer to keys of that relation in your answer, show how you know that key is indeed a key for that relation. [3]

7) Write a Java program to create any one of the relations you designed, insert any 3 rows in that relation, and then find and output any 1 of those rows. [5]

8) Instead of having a database like this for UCT alone, you are asked to design a database that can allow hundreds of universities across the world to share course content and course assessments (assignments/quizzes). For this repository, polyglot persistence (i.e. using a number of databases of different types for different aspects of the data) is now required.
   (a) Which part of the Amathuba data would be best suited to implementing as a document store when the data is on this worldwide scale – content or assessments? Why? Give an example JSON document that could be used for this part of the data.
   (b) Which part of the Amathuba data would be best suited to implementing as a key-value store when the data is on this worldwide scale – content or assessments? Why? State what the keys and values should be used for this part of the data. [4]

9) Briefly state each member's contribution to this assignment. [1]